

NUCLEOTIDE & NUCLEIC ACID CHEMISTRY

DR TANUSHREE DEBNATH

DEMONSTRATOR

DEPARTMENT OF BIOCHEMISTRY

GAUHATI MEDICAL COLLEGE & HOSPITAL

NUCLEIC ACIDS

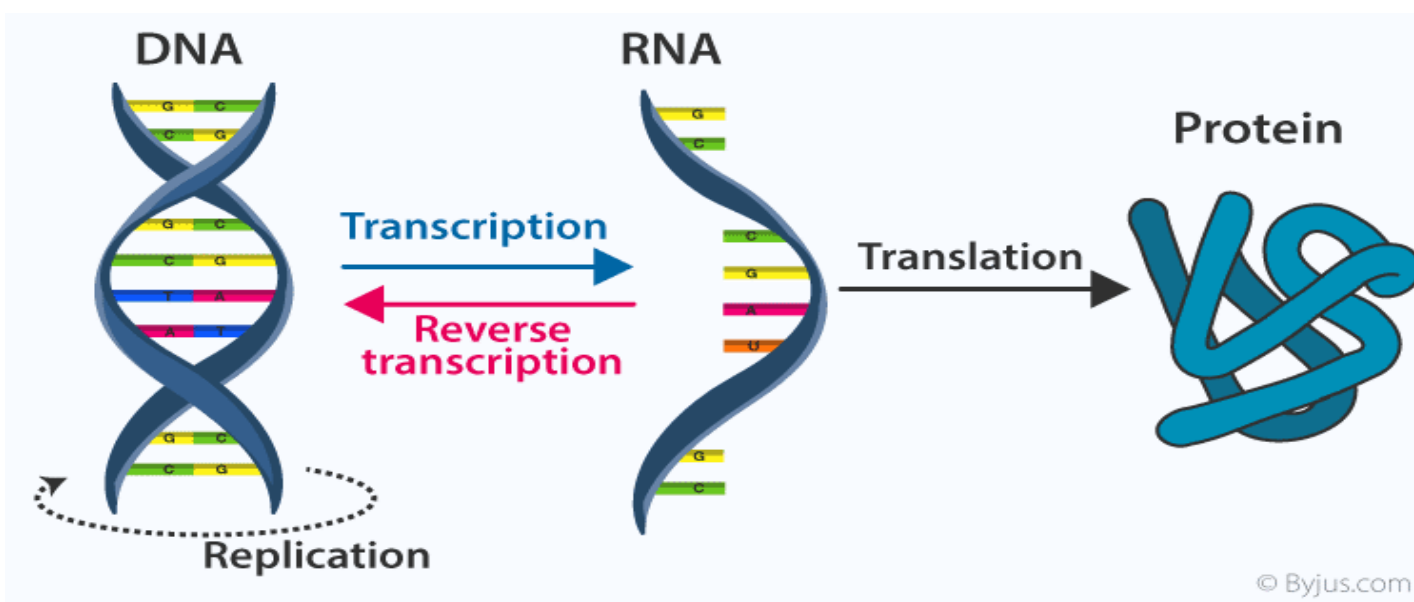
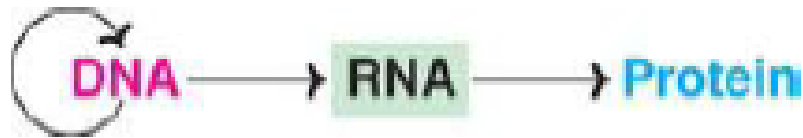
- *Nucleic acids are the polymers of nucleotides* (polynucleotides) held by 3' and 5' phosphate bridges.
- In other words, nucleic acids are built up by the monomeric units—nucleotides.
- There are two types of nucleic acids, namely
 - i) *Deoxyribonucleic acid* (DNA)
 - ii) *Ribonucleic acid* (RNA).
- Primarily, nucleic acids serve as repositories and transmitters of genetic information.

FUNCTIONS OF NUCLEIC ACIDS

- DNA is the chemical basis of heredity and may be regarded as the reserve bank of genetic information.
- DNA is exclusively responsible for maintaining the identity of different species of organisms over millions of years.
- Every aspect of cellular function is under the control of DNA.
- The ***DNA*** is organized into ***genes***, the fundamental units of ***genetic information***.
- The genes control the protein synthesis through the mediation of RNA

CENTRAL DOGMA OF LIFE

- The interrelationship of these three classes of biomolecules (DNA, RNA and proteins) constitutes the ***central dogma of molecular biology*** or more commonly the ***central dogma of life***.

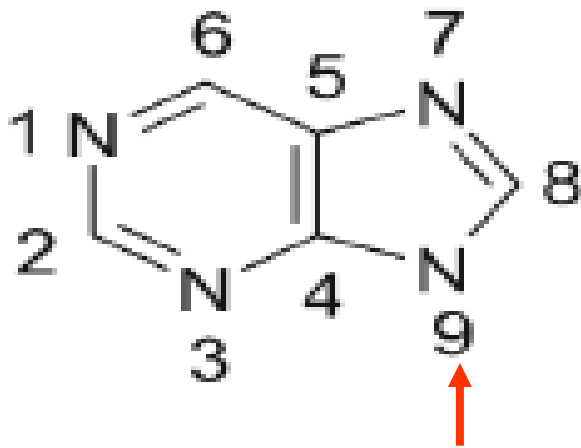


NUCLEOTIDES

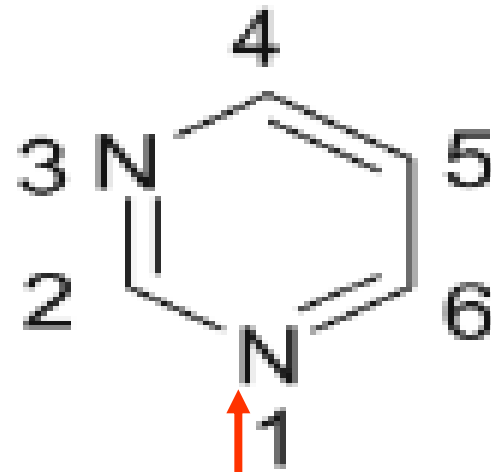
- Nucleotides are composed of :-
 - i) A *nitrogenous base*
 - ii) A *pentose sugar*
 - iii) A *phosphate*.
- The term **nucleoside** refers to **base + sugar**.
- Thus,
nucleotide = nucleoside + phosphate.

PURINES & PYRIMIDINES

- The nitrogenous bases found in nucleotides (and, therefore, nucleic acids) are ***aromatic heterocyclic compounds***.
- The bases are of two types—
 - Purines
 - pyrimidines.



PURINE RING



Pyrimidine

MAJOR BASES IN NUCLEIC ACIDS

○ PURINE

- i. Adenine
- ii. Guanine

○ PYRIMIDINE

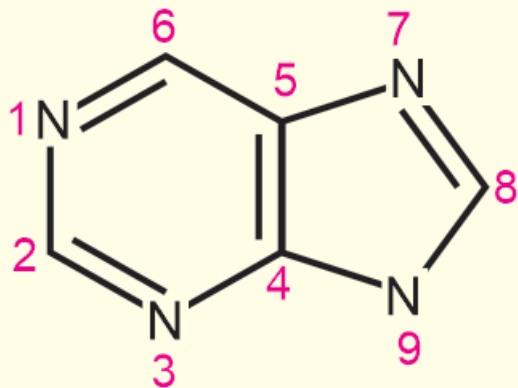
- i. Cytosine
- ii. Uracil
- iii. Thymine

THYMINE is the base
present in DNA

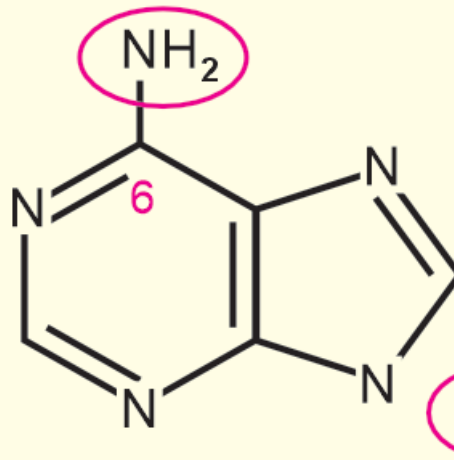
THIAMINE is a member of
vitamin B complex

- DNA and RNA contain the same purines namely
 - i. Adenine (A)
 - ii. Guanine (G).
- Further, the pyrimidine cytosine (C) is found in both DNA and RNA.
- However, the nucleic acids differ with respect to the second pyrimidine base.
- *DNA contains thymine (T) whereas RNA contains uracil (U).*

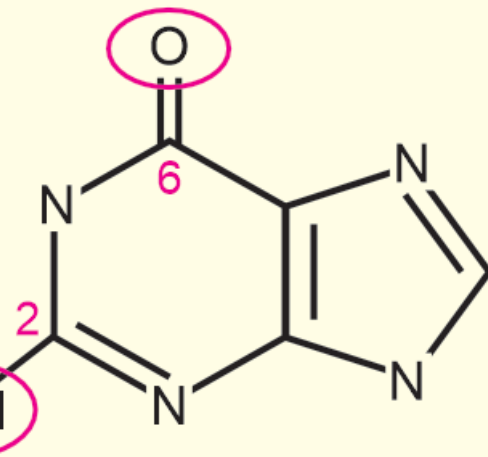
STRUCTURE OF PURINES



Purine ring

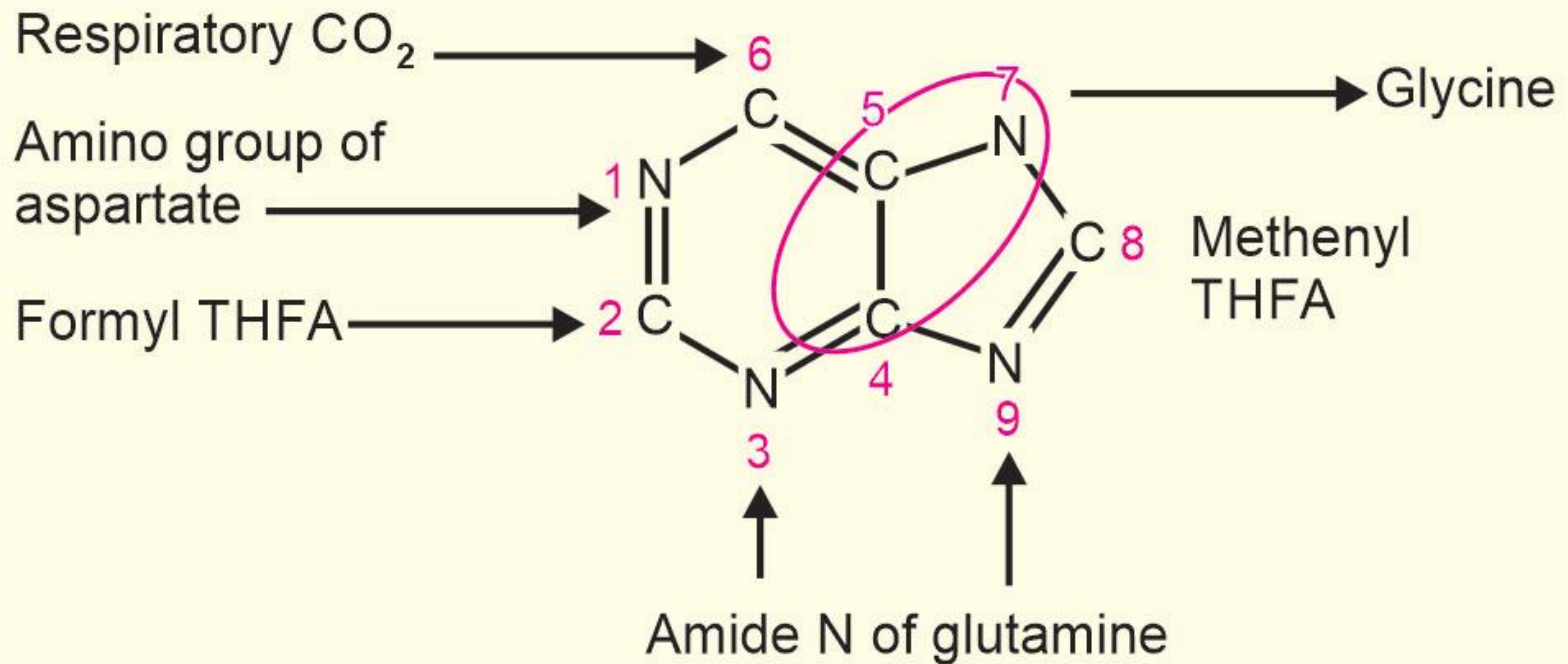


Adenine

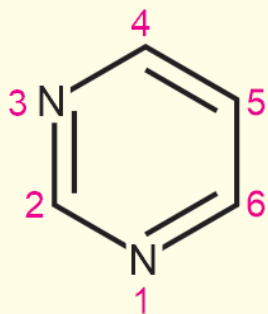


Guanine

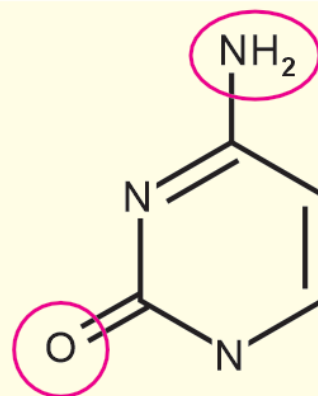
SOURCES OF THE PURINE RING



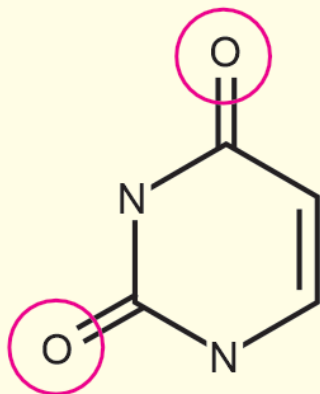
STRUCTURE OF PYRIMIDINES



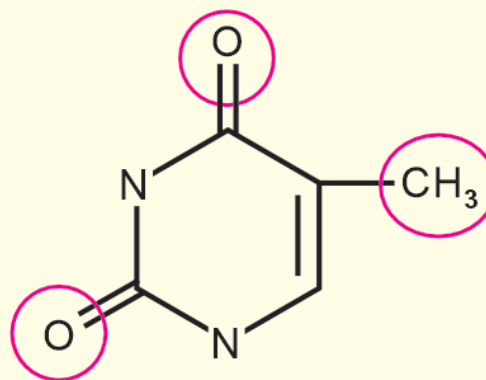
Pyrimidine



Cytosine

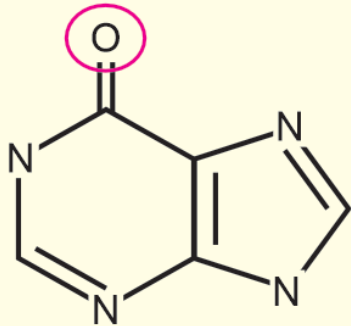


Uracil

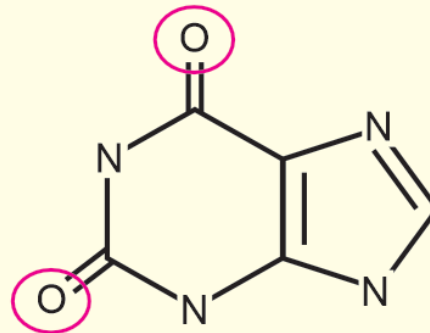


Thymine

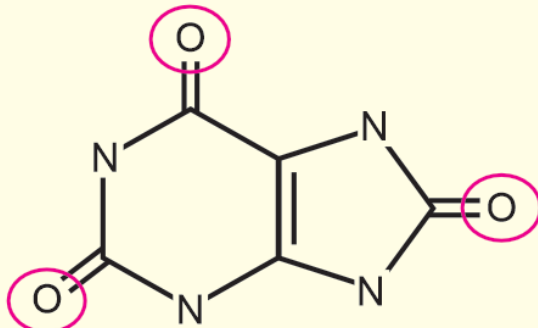
MINOR BASES SEEN IN NUCLEIC ACIDS



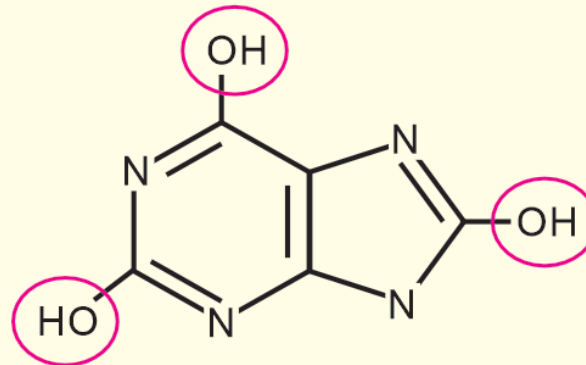
Hypoxanthine



Xanthine

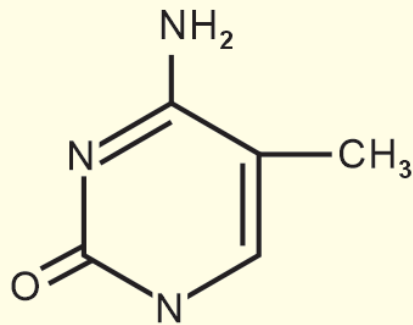


**Lactam or keto form
of uric acid**

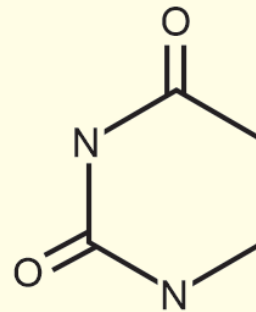


**Lactim or enol form
of uric acid**

MODIFIED PYRIMIDINE BASES



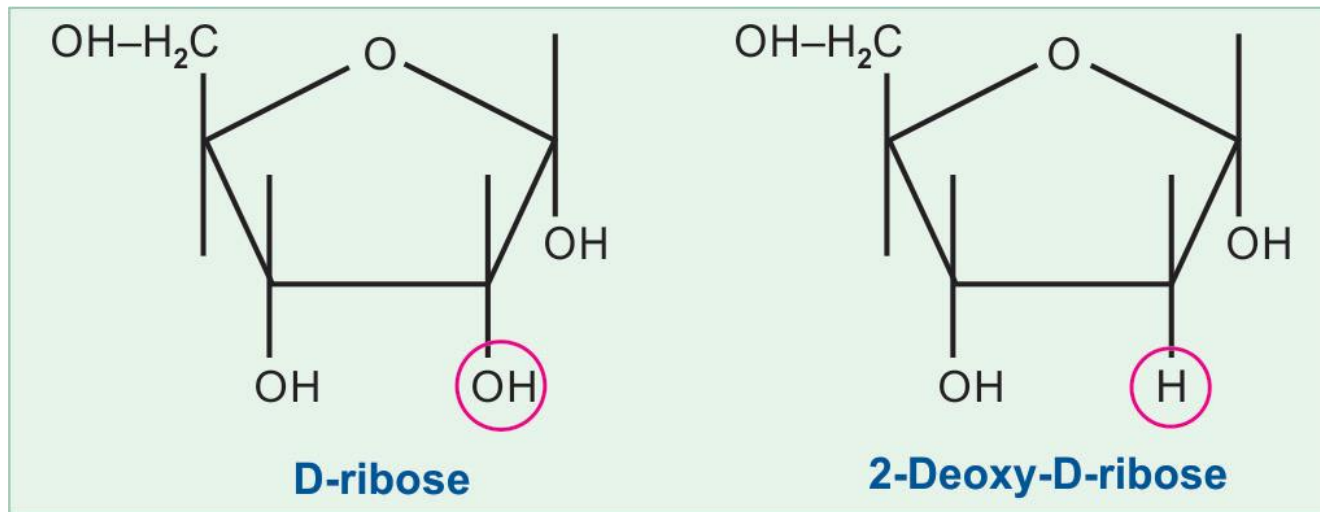
5-methyl cytosine



Dihydro-uracil

SUGARS OF NUCLEIC ACIDS

- The five carbon monosaccharides (pentoses) are found in the nucleic acid structure.
- **RNA** contains ***D-ribose*** while **DNA** contains ***D-deoxyribose***.
- Ribose and deoxyribose differ in structure at C2.
- Deoxyribose has one oxygen less at C2 compared to ribose



Sugar groups in nucleic acids

NOMENCLATURE OF NUCLEOTIDES

- The addition of a pentose sugar to base produces a nucleoside.
- If the sugar is ribose, ribonucleosides are formed.
- Adenosine, guanosine, cytidine and uridine are the ribonucleosides of A, G, C and U respectively.
- If the sugar is a deoxyribose, deoxyribonucleosides are produced.
- The pentoses are bound to nitrogenous bases by *β -N-glycosidic bonds*

Nucleosides

- **Base + sugar,**
(ribose or deoxy-ribose)
- **Bases are attached to the corresponding pentose sugar by**

N-glycosidic bond between
1st carbon of the pentose sugar
and
N9 of a purine
or
N1 of a pyrimidine.

Base + sugar = nucleosides

Ribonucleosides		
Adenine	+ Ribose	→ Adenosine
Guanine	+ Ribose	→ Guanosine
Uracil	+ Ribose	→ Uridine
Cytosine	+ Ribose	→ Cytidine
Hypoxanthine	+ Ribose	→ Inosine
Xanthine	+ Ribose	→ Xanthosine
Deoxyribonucleosides		
Adenine	+ Deoxy ribose	→Deoxy adenosine (d-adenosine)
Guanine	+ Deoxy ribose	→d-guanosine
Cytosine	+ Deoxy ribose	→d-cytidine
Thymine	+ Deoxy ribose	→d-thymidine

- The term mononucleotide is used when a single phosphate moiety is added to a nucleoside.
- Thus adenosine monophosphate (AMP) contains
AMP = adenine + ribose + phosphate.
- The addition of second or third phosphates to the nucleoside results in nucleoside diphosphate (e.g. ADP) or triphosphate (e.g. ATP), respectively.
- The anionic properties of nucleotides and nucleic acids are due to the negative charges contributed by phosphate groups.

Base+sugar+phosphate = nucleotide

Ribonucleotides		
Adenosine	+ Pi	Adenosine monophosphate (AMP) (Adenylic acid)
Guanosine	+ Pi	Guanosine monophosphate (GMP) (Guanylic acid)
Cytidine	+ Pi	Cytidine monophosphate (CMP) (Cytidylic acid)
Uridine	+ Pi	Uridine monophosphate (UMP) (Uridylic acid)
Inosine	+ Pi	Inosine monophosphate (IMP) (Inosinic acid)
Deoxyribonucleotides		
d-adenosine	+ Pi	d-AMP (d-adenylic acid)
d-guanosine	+ Pi	d-GMP (d-guanylic acid)
d-cytidine	+ Pi	d-CMP (d-cytidylic acid)
d-thymidine	+ Pi	d-TMP (d-thymidylic acid)

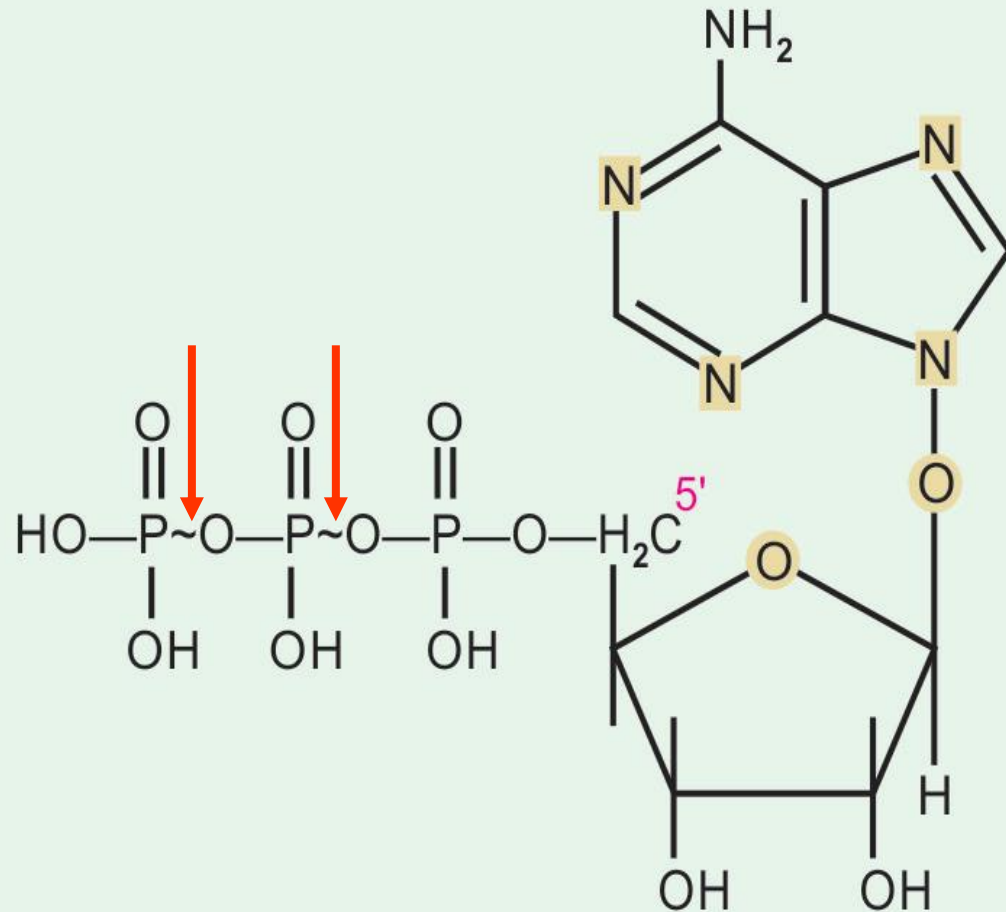
Nucleoside triphosphates

Nucleoside	Nucleoside monophosphate	Nucleoside diphosphate (NDP)	Nucleoside triphosphate (NTP)
Ribonucleoside phosphates			
Adenosine	Adenosine monophosphate (AMP)	Adenosine diphosphate (ADP)	Adenosine triphosphate (ATP)
Guanosine	GMP	GDP	GTP
Inosine	IMP	IDP	ITP
Cytidine	CMP	CDP	CTP
Uridine	UMP	UDP	UTP
Deoxyribonucleoside phosphates			
d-adenosine	d-AMP	d-ADP	d-ATP
d-guanosine	d-GMP	d-GDP	d-GTP
d-cytidine	d-CMP	d-CDP	d-CTP
d-thymidine	d-TMP	d-TDP	d-TTP

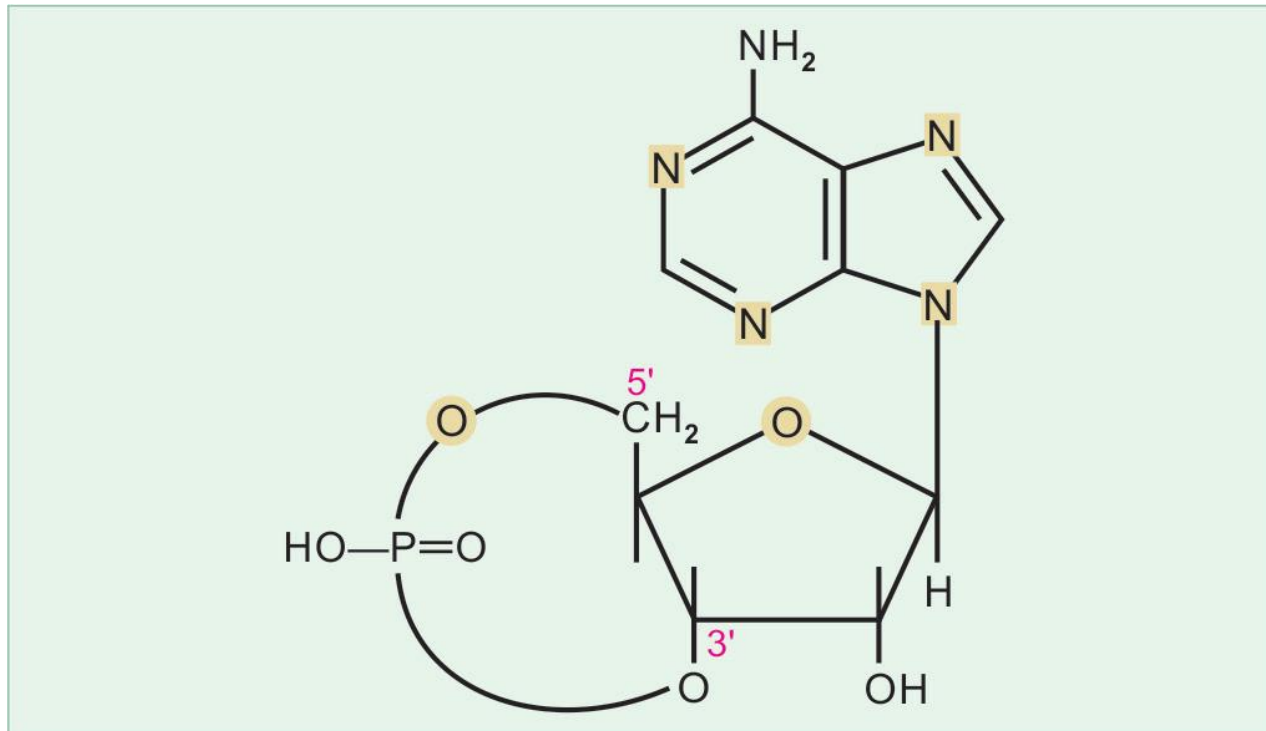
Nucleosides and nucleotides

Base	Sugar	Nucleoside	Phosphoric acid at	Nucleotide
Adenine	ribose	adenosine	5' position	AMP
do	do	do	3' position	3'-AMP
do	deoxyribose	d-adenosine	5' position	d-AMP
do	do	do	3' position	d-3'-AMP
Cytosine	ribose	cytidine	cytidine 5' position	CMP
do	do	do	3' position	3'-CMP
do	deoxyribose	d-cytidine	5' position	d-CMP
do	do	do	3' position	d-3'-CMP

Since 5'-nucleotides are more often seen, they are simply written without any prefix. For example, 5'-AMP is abbreviated as AMP; but 3' variety is always written as 3'-AMP.



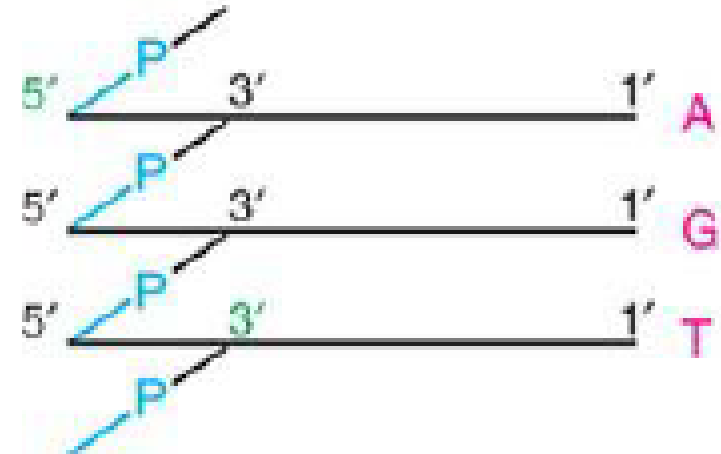
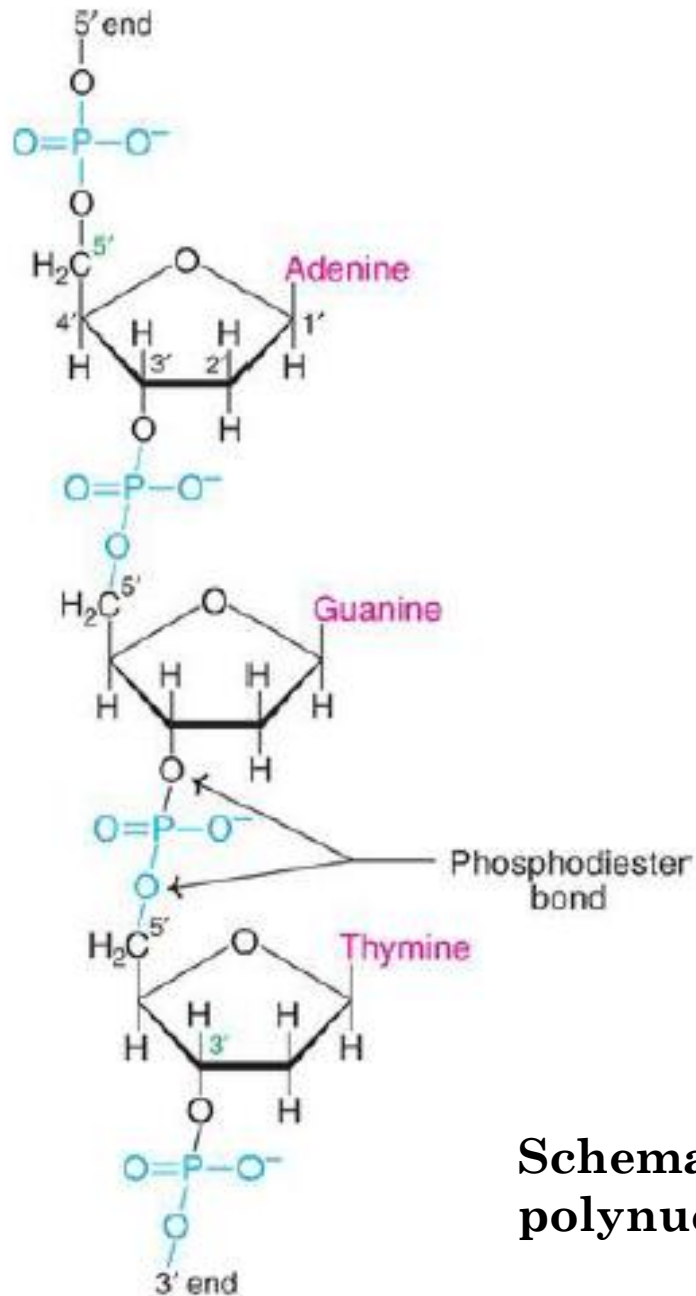
ATP is the universal energy currency.



3',5'-cyclic AMP or cAMP. A phosphodiester linkage is formed between the 3' and 5' positions of ribose group. Cyclic AMP and Cyclic GMP is a major metabolic regulators. These are second messengers in mediating the action of several hormones.

STRUCTURE OF DNA

- DNA is a ***polymer of deoxyribonucleotides*** (or simply deoxynucleotides).
- It is composed of monomeric units namely
 - i. deoxyadenylate (dAMP)
 - ii. deoxyguanylate (dGMP)
 - iii. deoxycytidylate (dCMP)
 - iv. deoxythymidylate (dTMP)

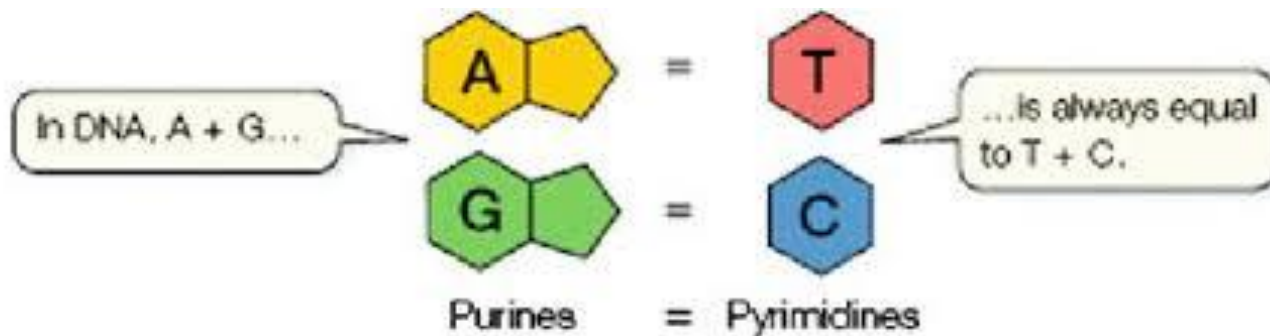


DNA (or RNA) structure is often represented in a short-hand form.

Schematic representation of polynucleotides

CHARGAFF'S RULE OF DNA COMPOSITION

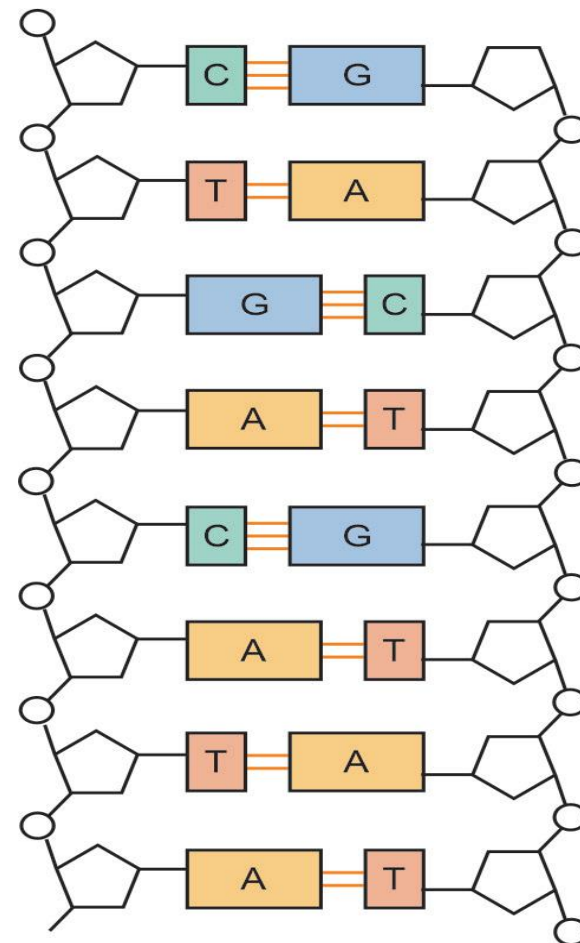
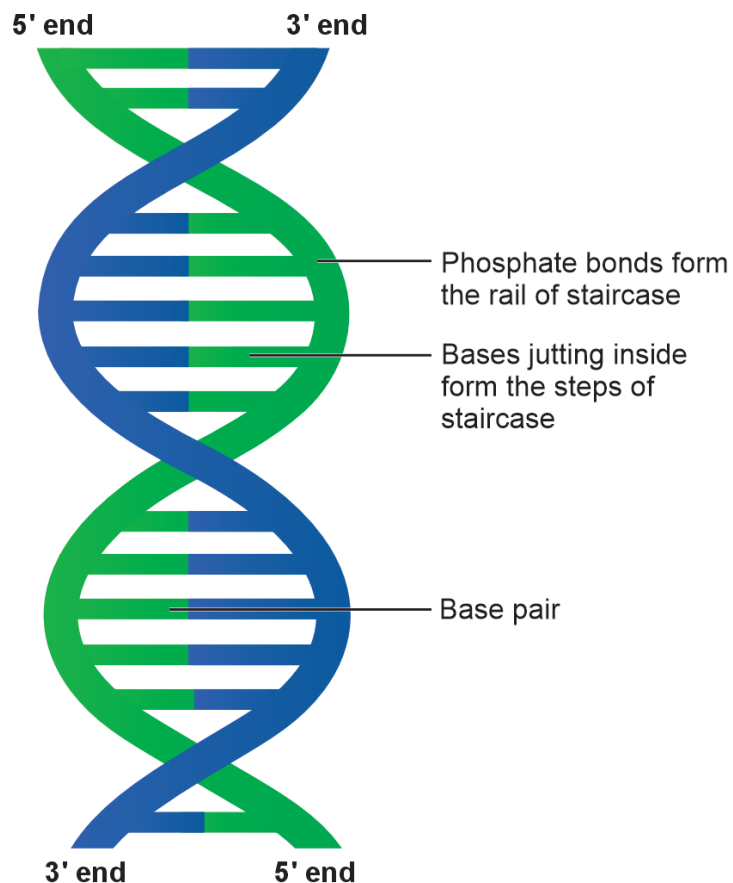
- Erwin Chargaff in late 1940s quantitatively analysed the DNA hydrolysates from different species.
-
- He observed that in all the species he studied, DNA had equal numbers of adenine and thymine residues ($A = T$) and equal numbers of guanine and cytosine residues ($G = C$).
- This is known as Chargaff's rule of *molar equivalence between the purines and pyrimidines in DNA* structure.
- Single stranded DNA, and RNAs which are usually single stranded, do not obey Chargaff's rule.

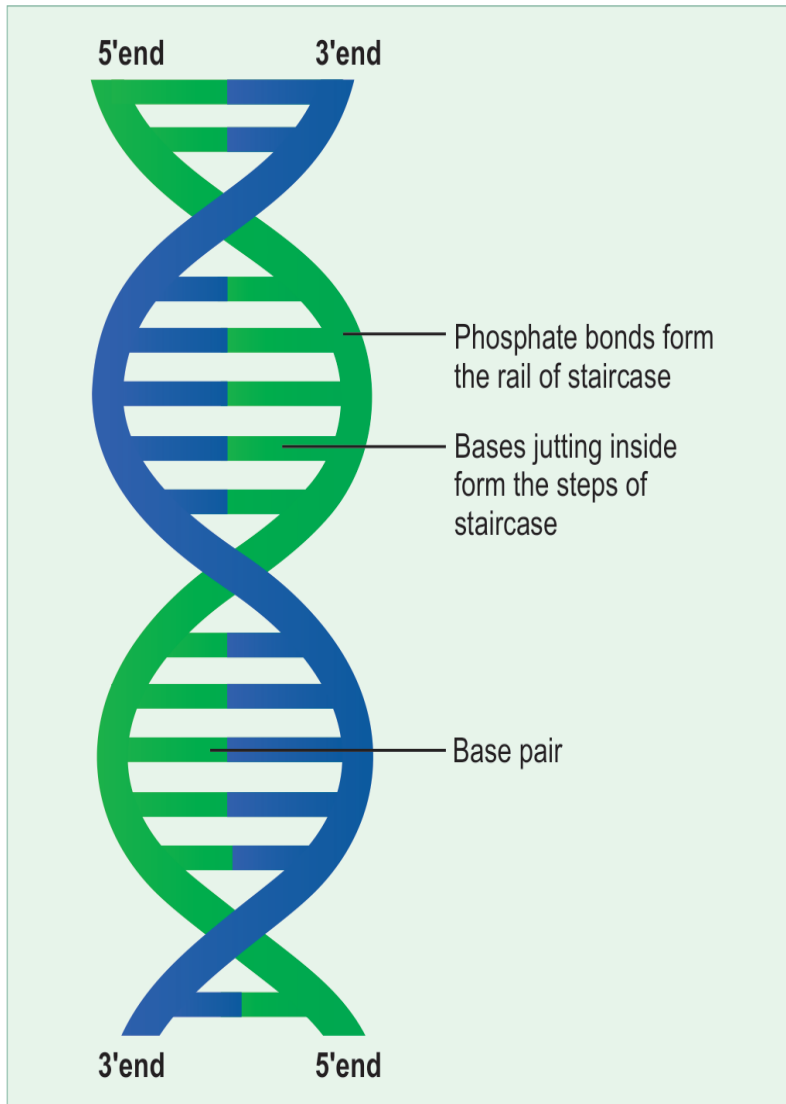


DNA DOUBLE HELIX

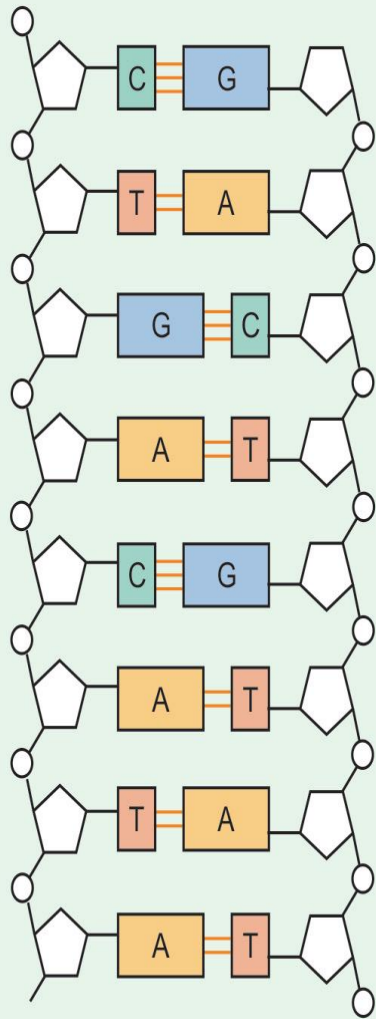
WATSON-CRICK MODEL OF DNA STRUCTURE

- The double helical structure of DNA was proposed by **James Watson** and **Francis Crick** in 1953 (Nobel Prize, 1962).





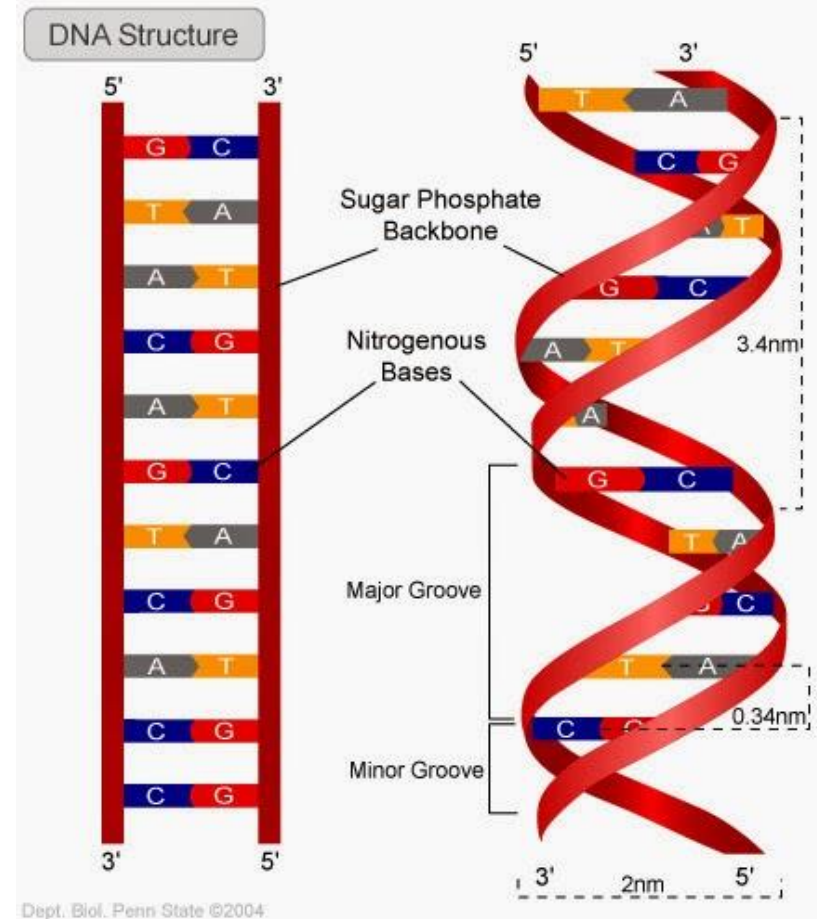
Watson–Crick model of double helical structure of DNA. Adjacent bases are separated by 0.34 nm. The diameter or width of the helix is 2 nanometers.



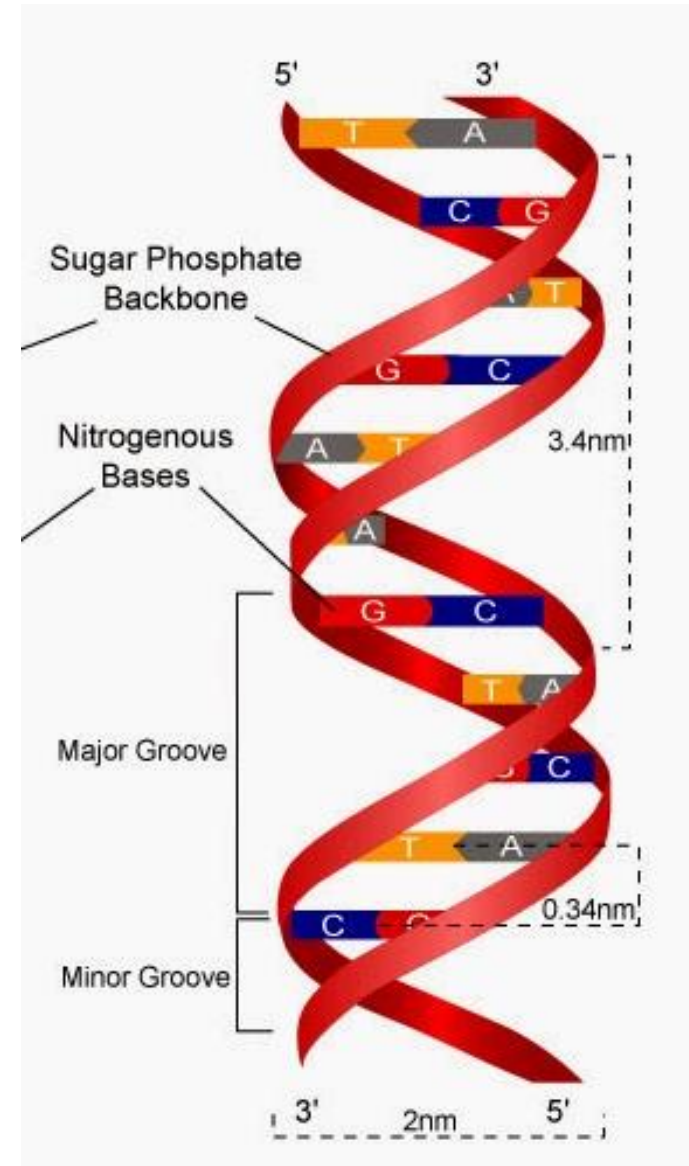
Base pairing rule. Base pairing of A with T and G with C. Hydrogen bonds keep the bases in position.

SALIENT FEATURES OF WATSON-CRICK MODEL OF DNA STRUCTURE

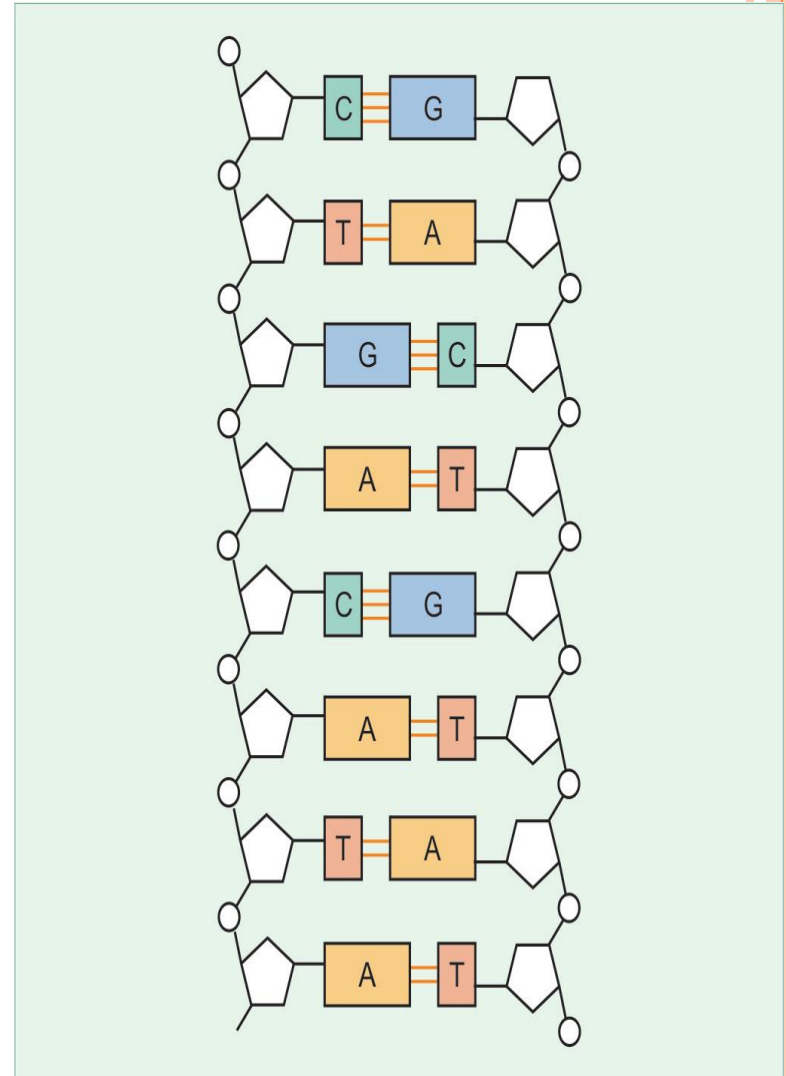
1. The DNA is a right handed double helix. It consists of *two polydeoxyribonucleotide chains* (strands) twisted around each other on a common axis.
2. The two strands are *antiparallel*, i.e., one strand runs in the 5'to 3'direction while the other in 3' to 5' direction. This is comparable to two parallel adjacent roads carrying traffic in opposite direction.
3. The width (or diameter) of a double helix is 20Å (2nm).



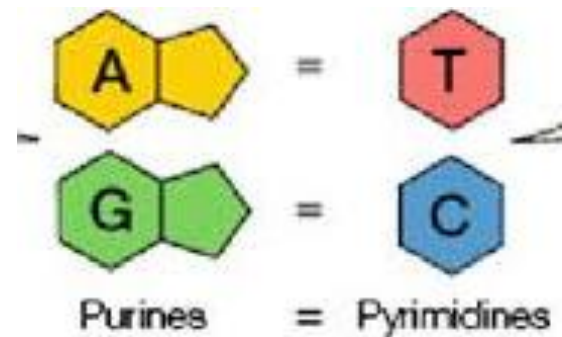
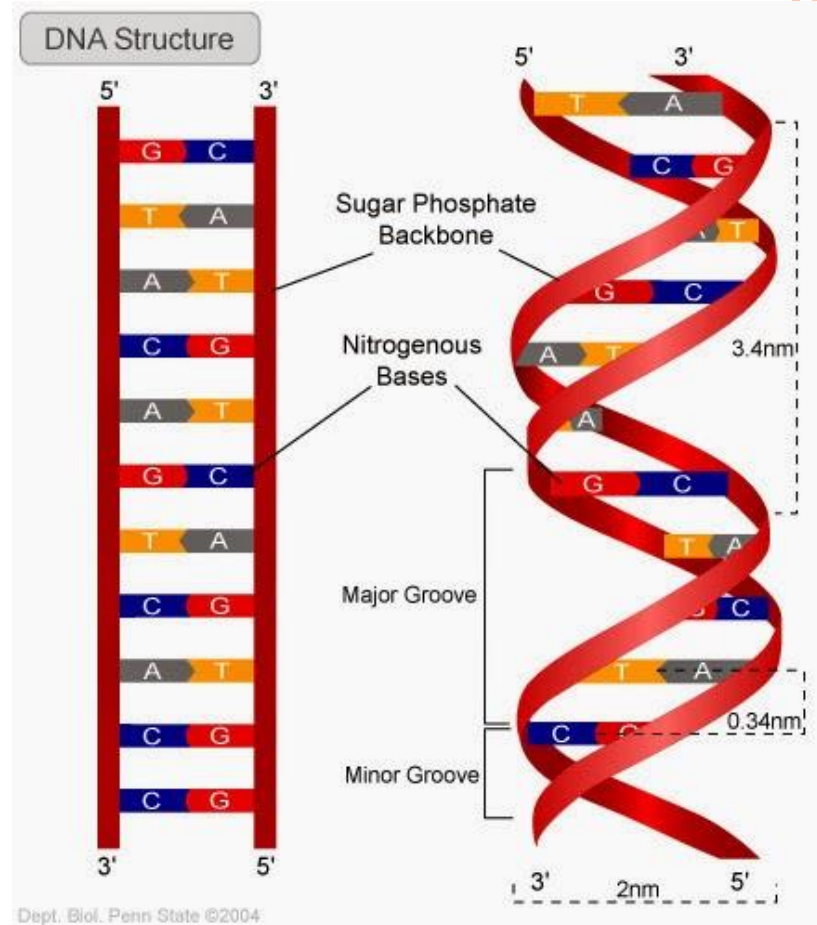
4. Each turn (pitch) of the helix is $34\text{\AA}$ (3.4nm) with **10 pairs of nucleotides**, each pair placed at a distance of about $3.4\text{\AA}$.
5. Each strand of DNA has a hydrophilic deoxyribose phosphate backbone ($3'$ - $5'$ phosphodiester bonds) on the outside (periphery) of the molecule while the hydrophobic bases are stacked inside (core).
6. The **two polynucleotide** chains are not identical but **complementary** to each other due to base pairing.

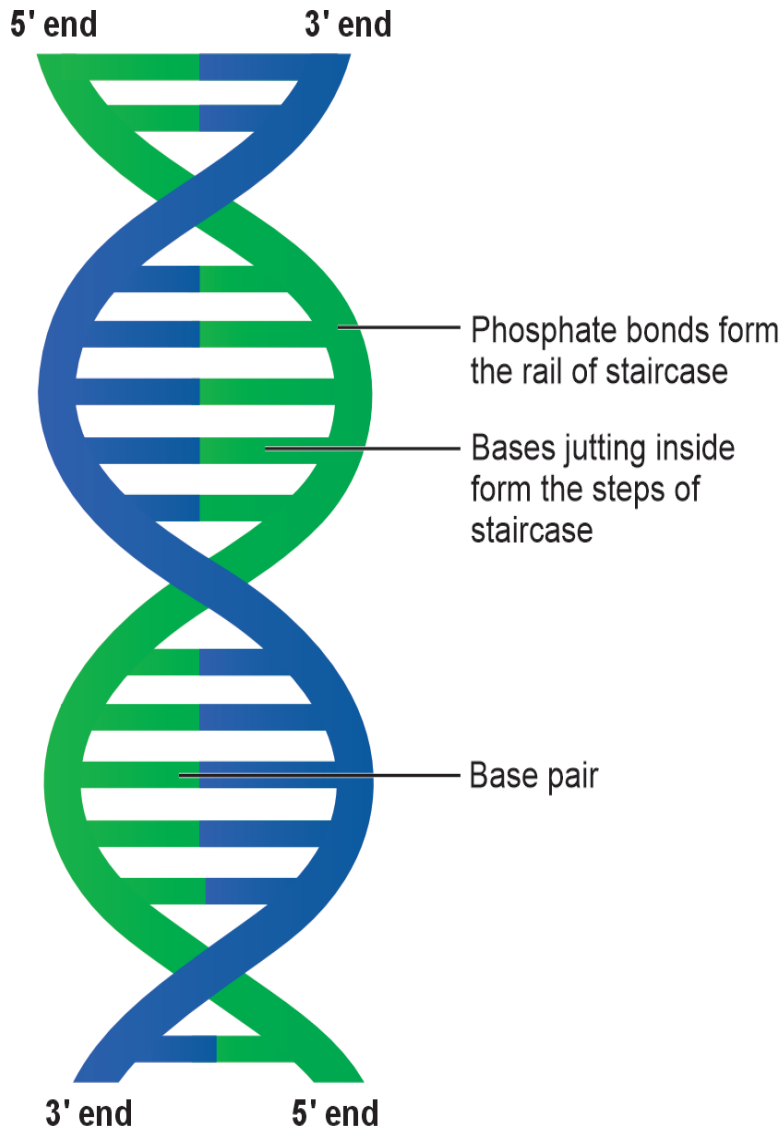


7. The two strands are ***held together by hydrogen bonds*** formed by complementary base pairs. The A-T pair has 2 hydrogen bonds while G-C pair has 3 hydrogen bonds. The G $\equiv$ C is stronger by about 50% than A = T.
8. The hydrogen bonds are formed between a purine and a pyrimidine only. If two purines face each other, they would not fit into the allowable space. And two pyrimidines would be too far to form hydrogen bonds. The only base arrangement possible in DNA structure, from spatial considerations is A-T, T-A, G-C and C-G.



9. The complementary base pairing in DNA helix proves ***Chargaff's rule***. The content of adenine equals to that of thymine ($A = T$) and guanine equals to that of cytosine ($G = C$).
10. The ***genetic information resides on*** one of the two strands known as ***template strand*** or sense strand. The opposite strand is antisense strand.
11. The double helix has (wide) major grooves and (narrow) minor grooves along the phosphodiester backbone. Proteins interact with DNA at these grooves, without disrupting the base pairs and double helix.





Watson-Crick Model of DNA Structure

- ❖ Right-handed double helix
- ❖ The base pairing rule
- ❖ Hydrogen bonding
- ❖ Antiparallel
- ❖ Size of the spiral
- ❖ Major and minor grooves

STRUCTURE OF RNA

- RNA is a polymer of ribonucleotides held together by 3',5'-phosphodiester bridges.
- Although RNA has certain similarities with DNA structure, they have specific differences:
 1. Pentose : The sugar in RNA is ribose in contrast to deoxyribose in DNA.
 2. Pyrimidine : RNA contains the pyrimidine uracil in place of thymine (in DNA).

- Single strand : RNA is usually a single stranded polynucleotide. However, this strand may fold at certain places to give a double stranded structure, if complementary base pairs are in close proximity.
- Chargaff's rule—not obeyed : Due to the single stranded nature, there is no specific relation between purine and pyrimidine contents. Thus the guanine content is not equal to cytosine (as is the case in DNA).

- Susceptibility to alkali hydrolysis : Alkali can hydrolyse RNA to 2',3'-cyclic diesters. This is possible due to the presence of a hydroxyl group at 2' position. DNA cannot be subjected to alkali hydrolysis due to lack of this group.
- Orcinol colour reaction : RNAs can be histologically identified by orcinol colour reaction due to the presence of ribose.

TYPES OF RNA:-

The three major types of RNAs with their respective cellular composition are given below:-

1. Messenger RNA (mRNA) : 5–10%
2. Transfer RNA (tRNA) : 10–20%
3. Ribosomal RNA (rRNA) : 50–80%

Other types of cellular RNA:-

1. Heterogeneous nuclear RNA (hnRNA)
2. Small nuclear RNA (snRNA)

FUNCTIONS OF RNA:-

Messenger RNA (mRNA)

1. Carriers genetic information from DNA to ribosomes.
2. Acts as a template for protein synthesis.
3. Formed during transcription.

Transfer RNA (tRNA)

1. Brings specific amino acids to the ribosome.
2. Has an anticodon that pairs with codon on mRNA.
3. Helps in assembling amino acids into polypeptide chain.

Ribosomal RNA (rRNA)

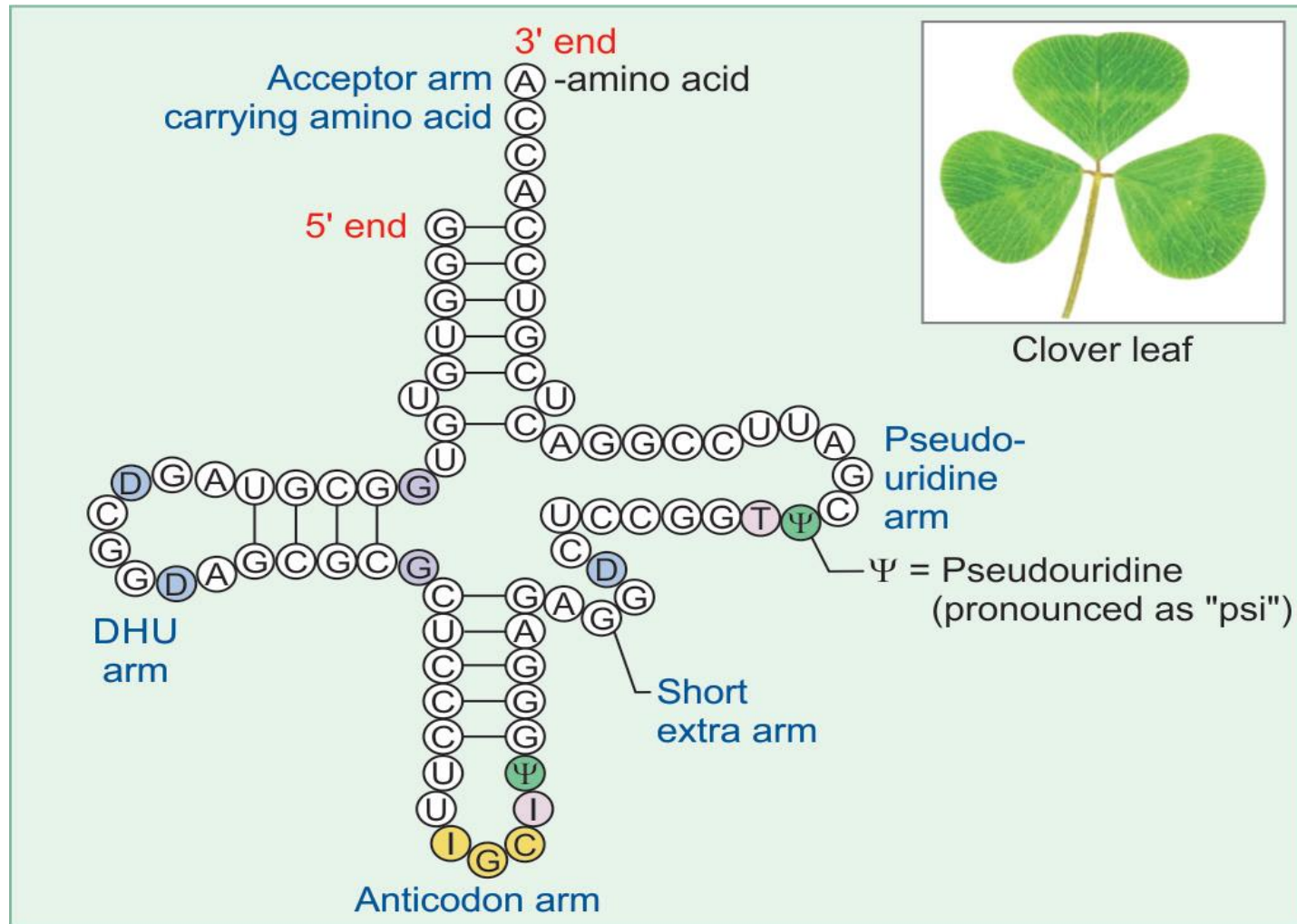
1. Structural and functional component of ribosomes.
2. Catalyzes peptide bond formation (acts as a ribozyme).
3. Forms the core of the ribosome.

CELLULAR RNAs & THEIR FUNCTIONS

Type of RNA	Abbreviation	Function(s)
Messenger RNA	mRNA	Transfers genetic information from genes to ribosomes to synthesize proteins.
Heterogeneous nuclear RNA	hnRNA	Serves as precursor for mRNA and other RNAs.
Transfer RNA	tRNA	Transfers amino acid to mRNA for protein biosynthesis.
Ribosomal RNA	rRNA	Provides structural framework for ribosomes.
Small nuclear RNA	snRNA	Involved in mRNA processing.
Small nucleolar RNA	snoRNA	Plays a key role in the processing of rRNA molecules.
Small cytoplasmic RNA	scRNA	Involved in the selection of proteins for export.
Transfer-messenger RNA	tmRNA	Mostly present in bacteria. Adds short peptide tags to proteins to facilitate the degradation of incorrectly synthesized proteins.

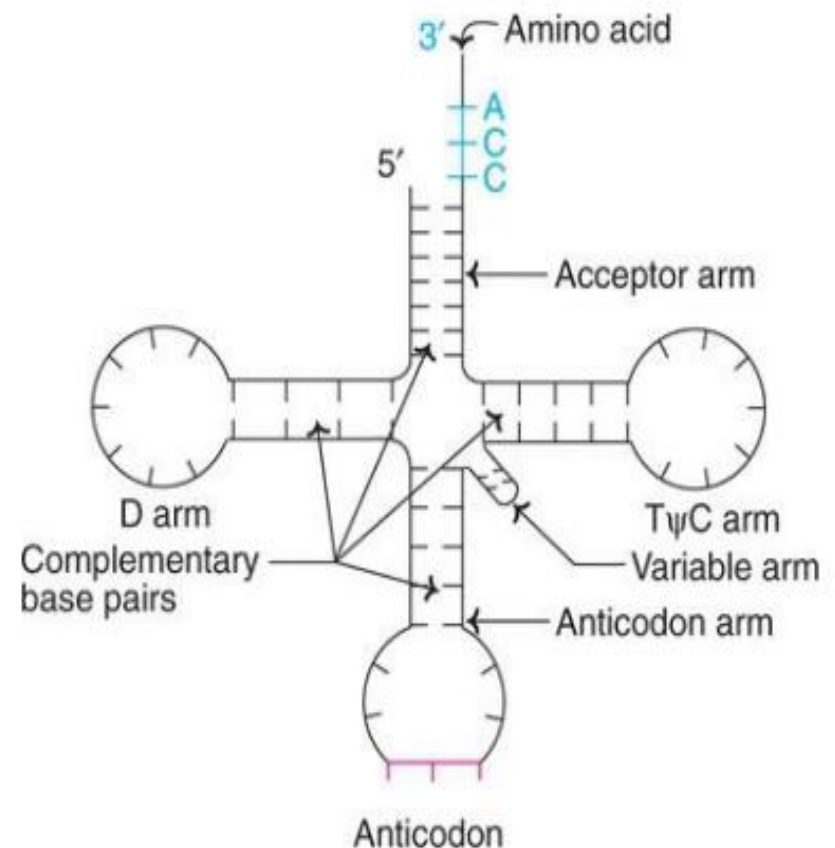
Transfer RNA (tRNA)

- Transfer RNA (soluble RNA) molecule contains 71–80 nucleotides (mostly 75) with a molecular weight of about 25,000.
- There are at least 20 species of tRNAs, corresponding to 20 amino acids present in protein structure.
- The structure of tRNA, resembles that of a clover leaf tRNA contains mainly four arms, each arm with a base paired stem.



Structure of Transfer RNA (tRNA)

1. The acceptor arm : This arm is capped with a sequence CCA (5'to 3'). The amino acid is attached to the acceptor arm.
2. The anticodon arm : This arm, with the three specific nucleotide bases (anticodon), is responsible for the recognition of triplet codon of mRNA. The codon and anticodon are complementary to each other.
3. The D arm : It is so named due to the presence of dihydrouridine.
4. The T ψ C arm : This arm contains a sequence of T, pseudouridine (represented by psi, ψ) and C.
5. The variable arm : This arm is the most variable in tRNA.



➤ Based on the variability of the variable arm, tRNAs are classified into 2 categories :-

(a) Class I tRNAs : The most predominant (about 75%) form with 3–5 base pairs length.

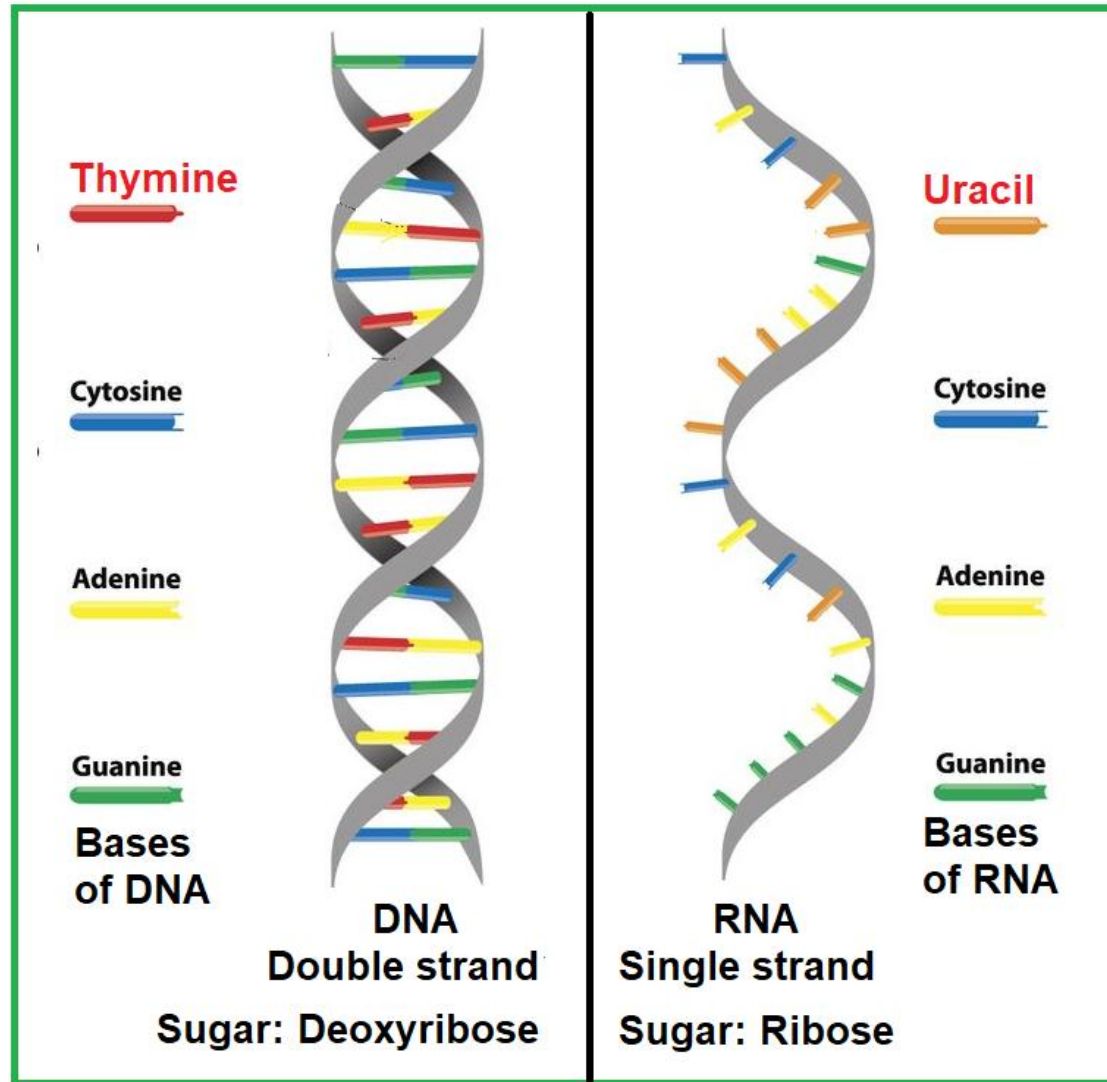
(b) Class II tRNAs : They contain 13–20 base pair long arm

➤ The structure of tRNA is maintained due to the complementary base pairing in the arms. The four arms with their respective base pairs are given below:-

- The acceptor arm – 7bp
- The T ψ C arm – 5bp
- The anticodon arm – 5bp
- The D arm – 4bp

DIFFERENCES BETWEEN RNA & DNA

RNA	DNA
1. Mainly seen in cytoplasm	Mostly inside nucleus
2. Usually 100-5000 bases	Millions of base pairs
3. Generally single stranded	Double stranded
4. Sugar is ribose	Sugar is deoxyribose
5. Purines: Adenine, Guanine Pyrimidines: Cytosine, Uracil	Adenine, Guanine Cytosine, Thymine
6. Guanine content is not equal to cytosine and adenine is not equal to uracil	Guanine is equal to cytosine and adenine is equal to thymine
7. Easily destroyed by alkali	Alkali resistant



THANK YOU